E-COMMERCE DATABASE MANAGEMENT SYSTEM

SALES AND CUSTOMER ANALYTICS SYSTEM

1.Aggregate functions

1.1 COUNT – Total customers

```
SQL> SELECT COUNT(*) AS TOTAL_CUSTOMERS
  2  FROM CUSTOMER;

TOTAL_CUSTOMERS
---------------
              5
```

1.2 SUM – Total sales

```
SQL> SELECT SUM(TOTAL_AMOUNT) AS TOTAL_SALES
  2  FROM ORDERS;

TOTAL_SALES
-----------
      11995
```

1.3 AVG – Average order amount

```
SQL> SELECT AVG(TOTAL_AMOUNT) AS AVERAGE_SALES
  2  FROM ORDERS;

AVERAGE_SALES
-------------
         2399
```

1.4 MIN – Minimum order amount

```
SQL> SELECT MIN(TOTAL_AMOUNT) AS MINIMUM_SALES
  2  FROM ORDERS;

MINIMUM_SALES
-------------
         1499
```

1.5 MAX – Maximum order amount

```
SQL>  SELECT MAX(TOTAL_AMOUNT) AS MAXIMUM_SALES
  2   FROM  ORDERS;

MAXIMUM_SALES
-------------
         3299
```

2. Total sales report

```
SQL> SELECT
  2      COUNT(*) AS TOTAL_ORDERS,
  3      SUM(TOTAL_AMOUNT) AS TOTAL_SALES
  4  FROM ORDERS;

TOTAL_ORDERS TOTAL_SALES
------------ -----------
           5       11995
```

3. Top customers based on purchase amount

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME,
  4      SUM(O.TOTAL_AMOUNT) AS TOTAL_PURCHASE
  5  FROM CUSTOMER C
  6  JOIN ORDERS O
  7  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  8  GROUP BY C.CUSTOMER_ID, C.FIRST_NAME
  9  ORDER BY TOTAL_PURCHASE DESC;

CUSTOMER_ID FIRST_NAME                     TOTAL_PURCHASE
----------- ------------------------------ --------------
          3 Ananya                                   3299
          5 Priya                                    2899
          1 Pooja                                    2499
          2 Rahul                                    1799
          4 Arjun                                    1499
```

4. Best-selling products

```
SQL> SELECT
  2      P.PRODUCT_ID,
  3      P.PRODUCT_NAME,
  4      SUM(O.QUANTITY) AS TOTAL_SOLD
  5  FROM PRODUCT P
  6  JOIN ORDERS O
  7  ON P.PRODUCT_ID = O.PRODUCT_ID
  8  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  9  ORDER BY TOTAL_SOLD DESC
 10  FETCH FIRST 1 ROW ONLY;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
TOTAL_SOLD
----------
       103
Running Shoes
         3
```

5. Category-wise sales analysis

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME,
  4      SUM(O.TOTAL_AMOUNT) AS TOTAL_PURCHASE
  5  FROM CUSTOMER C
  6  JOIN ORDERS O
  7  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  8  GROUP BY C.CUSTOMER_ID, C.FIRST_NAME
  9  ORDER BY TOTAL_PURCHASE DESC;

CUSTOMER_ID FIRST_NAME                     TOTAL_PURCHASE
----------- ------------------------------ --------------
          3 Ananya                                   3299
          5 Priya                                    2899
          1 Pooja                                    2499
          2 Rahul                                    1799
          4 Arjun                                    1499
```